

MABUCHI GEOMETRY RELATIVE TO $\mathcal{I}$ -MODEL POTENTIALS

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ABSTRACT. We construct a complete geodesic metric $d_{p,\phi}$ on the relative finite energy space $\mathcal{E}^p(X, \theta, \phi)$ for every $p \geq 1$, where θ represents a big cohomology class and ϕ is an $\mathcal{I}$ -model potential of positive mass. The construction extends Gupta's approximation method by establishing continuity of relative envelopes under d_S -approximation of singularity types.

1. INTRODUCTION

On a compact Kähler manifold (X, ω_X) , the problem of finding the canonical metric is closely related to studying the geometric structure on the *space of Kähler potentials*

$$\mathcal{H}_{\omega_X} := \{\varphi \in C^\infty(X) : \omega_\varphi := \omega_X + dd^c \varphi > 0\},$$

which, up to normalization, parametrizes all Kähler metrics cohomologous to ω_X . In [Mab87], [Sem92], and [Don99], the authors discovered a Riemannian structure on $\mathcal{H}_{\omega_X}$ whose geodesic equation is a homogeneous complex Monge-Ampère equation in one extra dimension. In [Che00], Chen proved that this Riemannian structure gives rise to an honest metric d_2 on $\mathcal{H}_{\omega_X}$.

Later, Darvas [Dar15] identified $\mathcal{E}^p(X, \omega_X)$ with the metric completion of the space of smooth Kähler potentials for the L^p path-length distance. These completions are geodesic metric spaces, and their weak geodesics are constant-speed metric geodesics.

This picture extends beyond Kähler classes. Darvas–Di Nezza–Lu [DDL18a] constructed the complete geodesic metric d_1 in arbitrary big classes, and Di Nezza–Lu [DL20] constructed d_p for $p \geq 1$ in big and nef classes. Gupta [Gup25, Theorem 1.1] subsequently established the complete geodesic metric structure on $\mathcal{E}^p(X, \theta)$ for every big class $\{\theta\}$.

For potentials with prescribed singularities, the natural spaces are the relative finite energy classes $\mathcal{E}^p(X, \theta, \phi)$, where ϕ is a positive-mass model potential. In a Kähler class, Trusiani [Tru22] constructed a complete metric on $\mathcal{E}^1(X, \theta, \phi)$ using the relative Monge-Ampère energy. Xia [Xia23] constructed a locally complete metric on the relative $\mathcal{E}^p$ spaces by a pluripotential-theoretic approach. Gupta [Gup23] established a complete metrizable topology on relative low energy spaces. For model potentials of analytic singularity type, Gupta [Gup25, Theorem 3.7] further constructed complete metrics on $\mathcal{E}^p(X, \theta, \phi)$ for which weak geodesics are constant-speed metric geodesics.

Our main result extends Gupta's construction [Gup25, Theorem 3.7] from analytic singularity types to positive-mass $\mathcal{I}$ -model potentials. The notation is recalled below.

Theorem 1.1. *Let (X, ω_X) be a compact Kähler manifold of dimension n , and let θ be a smooth closed real $(1, 1)$ -form representing a big class. Let $\phi \in \text{PSH}(X, \theta)$ satisfy*

$$(1.2) \quad \phi = P_\theta[\phi] = P_\theta[\phi]_{\mathcal{I}}, \quad m_\phi := \int_X \theta_\phi^n > 0.$$

For every $p \geq 1$, there is a complete metric $d_{p,\phi}$ on $\mathcal{E}^p(X, \theta, \phi)$ for which weak geodesics are constant-speed metric geodesics. When $p = 1$, $d_{1,\phi}$ is given by the following formula, which agrees with the metric in [Tru22]

in the Kähler setting:

$$(1.3) \quad d_{1,\phi}(u, v) = E_{\theta,\phi}(u) + E_{\theta,\phi}(v) - 2E_{\theta,\phi}(P_\theta(u, v)).$$

Moreover, recall the I_p -energy on $\mathcal{E}^p(X, \theta, \phi)$:

$$I_p(u, v) := \int_X |u - v|^p (\theta_u^n + \theta_v^n), \quad u, v \in \mathcal{E}^p(X, \theta, \phi).$$

Then $d_{p,\phi}$ is equivalent to the I_p -energy, i.e., there exists $C = C(n, p) > 0$ such that

$$C^{-1}I_p(u, v) \leq d_{p,\phi}(u, v)^p \leq CI_p(u, v).$$

It is also bi-Lipschitz equivalent to Xia's metric (Corollary 4.5). Consequently, the two metrics have the same Cauchy sequences and limits, and Xia's metric is complete in this case.

The construction follows Gupta's approximation method. By [DX24, Theorem 3.8], the $\mathcal{I}$ -model hypothesis provides analytic singularity types converging to $[\phi]$ in $d_{\mathcal{S}}$. These approximations need not be monotone, so the passage to the limit requires an additional continuity argument. After extracting a suitable sequence (ϕ_j) , we prove that relative envelopes of $C^{1,\bar{1}}$ obstacles converge in capacity and that their Monge–Ampère measures converge in total variation.

For $u_i = P_\theta[\phi](f_i)$ with $f_i \in C^{1,\bar{1}}(X)$, this continuity gives the geodesic distance formula

$$\lim_{j \rightarrow \infty} d_{p,\phi_j}(P_\theta[\phi_j](f_0), P_\theta[\phi_j](f_1))^p = \int_X |\dot{u}_0|^p \theta_{u_0}^n,$$

where (u_t) is the weak geodesic joining u_0 and u_1 . The geodesic distance formula makes the limiting distance independent of the approximating sequence and the obstacle representatives. Decreasing approximation then extends the metric to $\mathcal{E}^p(X, \theta, \phi)$.

AI Disclosure. This note grew out of the author's investigation of a different problem. The author and his advisor, Tamás Darvas, had previously considered the possibility of extending Gupta's construction in [Gup25] to $\mathcal{I}$ -model potentials through approximation in the $d_{\mathcal{S}}$ -metric by potentials of analytic singularity type. The author used GPT 6 Astra to help develop and revise the detailed arguments, identify gaps, and check citations. The author subsequently checked, refined, and rewrote the arguments. GPT 6 Astra also assisted with language editing.

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2. NOTATION AND SETUP

Let (X, ω_X) be a compact Kähler manifold of dimension n , and let θ be a smooth closed real $(1, 1)$ -form representing a big class. We write $\text{PSH}(X, \theta)$ for the set of θ -psh functions and

$$\theta_u^n := \langle (\theta + dd^c u)^n \rangle$$

for the non-pluripolar Monge–Ampère measure of $u \in \text{PSH}(X, \theta)$. Mixed products are also understood in the non-pluripolar sense. We refer to [DDL25] for more details on the relative pluripotential theory used below.

For a function f on X , we define the *Monge–Ampère envelope* by

$$P_\theta(f) := (\sup\{u \in \text{PSH}(X, \theta) : u \leq f\})^*, \quad P_\theta(f, g) := P_\theta(\min\{f, g\}),$$

where $*$ denotes upper semicontinuous regularization. For $\psi \in \text{PSH}(X, \theta)$, set

$$P_\theta[\psi](f) := \left(\lim_{C \rightarrow \infty} P_\theta(\psi + C, f) \right)^*, \quad P_\theta[\psi] := P_\theta[\psi](0).$$

A potential ϕ is *model* if $\phi = P_\theta[\phi]$. We write $u \leq v$ if $u \leq v + C$ for some constant C , and $[u] = [v]$ if both $u \leq v$ and $v \leq u$.

Let $\mathcal{I}(t\psi)$ denote the multiplier ideal sheaf of $t\psi$. Following [DX24], we define the $\mathcal{I}$ -envelope:

$$P_\theta[\psi]_{\mathcal{I}} := \sup \{v \in \text{PSH}(X, \theta) : v \leq 0, \mathcal{I}(tv) \subseteq \mathcal{I}(t\psi) \text{ for every } t > 0\}.$$

Throughout the note, we fix a potential ϕ satisfying

$$(2.1) \quad \phi = P_\theta[\phi] = P_\theta[\phi]_{\mathcal{I}}, \quad m_\phi := \int_X \theta_\phi^n > 0.$$

The relative full-mass and finite-energy classes are

$$\mathcal{E}(X, \theta, \phi) := \left\{ u \in \text{PSH}(X, \theta) : u \leq \phi, \int_X \theta_u^n = m_\phi \right\}$$

and, for $p \geq 1$,

$$\mathcal{E}^p(X, \theta, \phi) := \left\{ u \in \mathcal{E}(X, \theta, \phi) : \int_X |u - \phi|^p \theta_u^n < \infty \right\}.$$

We use the comparison functional

$$I_p(u, v) := \int_X |u - v|^p (\theta_u^n + \theta_v^n)$$

and the class of special envelopes:

$$\mathcal{H}_{\theta, \phi} := \{P_\theta[\phi](f) : f \in C^{1,1}(X)\},$$

where $C^{1,1}(X)$ denotes the continuous functions with bounded distributional Laplacian. Every element of $\mathcal{E}^p(X, \theta, \phi)$ admits a decreasing approximation by elements of $\mathcal{H}_{\theta, \phi}$.

For u with the same singularity type as ϕ , the relative Monge–Ampère energy is

$$E_{\theta, \phi}(u) := \frac{1}{n+1} \sum_{k=0}^n \int_X (u - \phi) \theta_u^k \wedge \theta_\phi^{n-k}.$$

It extends to $\mathcal{E}^1(X, \theta, \phi)$ by decreasing approximation. We also write

$$P_\theta[\phi](u, v) := P_\theta[\phi](\min\{u, v\}).$$

For $u, v \in \mathcal{E}^p(X, \theta, \phi)$, this agrees with $P_\theta(u, v)$.

We use the envelope construction of weak geodesics described in [DDL18b, Section 3.1]. Let $D := \{\zeta \in \mathbb{C} : 1 < |\zeta| < e\}$ and let $\pi : X \times D \rightarrow X$ be the projection. A curve $(v_t)_{0 < t < 1}$ in $\text{PSH}(X, \theta)$ is a subgeodesic if

$$(x, \zeta) \longmapsto v_{\log |\zeta|}(x) \in \text{PSH}(X \times D, \pi^* \theta).$$

For $u_0, u_1 \in \mathcal{E}^p(X, \theta, \phi)$, their weak geodesic u_t is the upper semicontinuous regularization of the supremum of all such subgeodesics satisfying

$$\limsup_{t \rightarrow 0} v_t \leq u_0, \quad \limsup_{t \rightarrow 1} v_t \leq u_1 \quad \text{pointwise on } X.$$

We denote its right derivative at $t = 0$ by $\dot{u}_0$.

Throughout, convergence in capacity means convergence with respect to the Bedford–Taylor capacity Cap_{ω_X} , and $|\nu|$ denotes the total variation measure of a signed measure ν .

3. APPROXIMATION BY POTENTIALS WITH ANALYTIC SINGULARITIES

As indicated in the introduction, we now construct the approximation sequence and establish the envelope convergence needed in the construction.

Since ϕ is an $\mathcal{I}$ -model potential, [DX24, Lemma 3.2, Theorem 3.8] provide model potentials $\phi_j = P_\theta[\phi_j] \in \text{PSH}(X, \theta)$ of analytic singularity type converging in the d_S -metric:

$$d_S([\phi_j], [\phi]) \longrightarrow 0.$$

By [DDL21, Lemma 3.7], we know that the masses of ϕ_j converge to the mass of ϕ :

$$m_{\phi_j} \longrightarrow m_\phi > 0.$$

After discarding finitely many terms, we may assume that $m_{\phi_j}, m_\phi \geq \delta$ for some $\delta > 0$.

Now, we apply [DDL21, Theorem 5.6]: after passing to a subsequence and relabeling, we find $v_j, w_j \in \text{PSH}(X, \theta)$ that sandwich ϕ_j :

$$(3.1) \quad w_j \leq \phi_j \leq v_j, \quad w_j \nearrow \phi \text{ almost everywhere}, \quad v_j \searrow \phi.$$

The proof there also gives $P_\theta[v_j] \searrow \phi$. We may therefore replace v_j by $P_\theta[v_j]$ and assume that v_j are model potentials. We will fix the sequence ϕ_j in the construction.

For $u_i = P_\theta[\phi](f_i) \in \mathcal{H}_{\theta, \phi}$, where $f_i \in C^{1,1}(X)$ and $i = 0, 1$, we set

$$u_i^j := P_\theta[\phi_j](f_i), \quad i = 0, 1.$$

We will define the metric on $\mathcal{H}_{\theta, \phi}$ by

$$(3.2) \quad d_{p, \phi}(u_0, u_1) := \lim_{j \rightarrow \infty} d_{p, \phi_j}(u_0^j, u_1^j),$$

where d_{p, ϕ_j} is the metric constructed in [Gup25, Theorem 3.7]. To justify this definition, we prove that the limit exists and is independent of both the obstacle representatives f_i and the chosen approximation ϕ_j by showing the geodesic distance formula

$$d_{p, \phi}(u_0, u_1)^p = \int_X |\dot{u}_0|^p \theta_{u_0}^n,$$

where u_t is the weak geodesic joining u_0 and u_1 .

Lemma 3.3. *Let (ϕ_j) be the sequence fixed above. For $h \in C^{1,1}(X)$, set*

$$V_j := P_\theta[\phi_j](h), \quad V := P_\theta[\phi](h).$$

Then $V_j \rightarrow V$ in capacity and their non-pluripolar measures converge in total variation:

$$|\theta_{V_j}^n - \theta_V^n|(X) \longrightarrow 0.$$

Proof. With the sandwiching sequence (3.1), we have:

$$a_j := P_\theta[w_j](h) \leq V_j \leq P_\theta[v_j](h) =: b_j.$$

As $w_j \nearrow \phi$, we can apply [Gup25, Lemma 4.2] so that

$$\left(\lim_j P_\theta[w_j](h) \right)^* = P_\theta[\phi](h) = V.$$

Thus $P_\theta[w_j](h) \rightarrow V$ in capacity. Meanwhile, $P_\theta[v_j](h)$ decreases to a θ -psh function $b \geq V$. As v_j is a model potential,

$$b_j \leq v_j + \sup_X h, \quad P_\theta[v_j](h) \leq h.$$

Passing to the limit gives $b \leq \phi + \sup_X h$ and $b \leq h$. Consequently,

$$b \leq P_\theta(\phi + \sup_X h, h) \leq P_\theta[\phi](h) = V.$$

Therefore $b = V$, and $P_\theta[v_j](h) \rightarrow V$ in capacity. For every $\varepsilon > 0$,

$$\{|V_j - V| > \varepsilon\} \subset \{|a_j - V| > \varepsilon\} \cup \{|b_j - V| > \varepsilon\}.$$

Subadditivity of capacity proves $V_j \rightarrow V$ in capacity.

For the second assertion, we denote the contact sets $D_j := \{V_j = h\}$ and $D := \{V = h\}$. By the work of Di Nezza–Trapani [DT21], we have

$$\theta_{V_j}^n = \mathbf{1}_{D_j} \theta_h^n, \quad \theta_V^n = \mathbf{1}_D \theta_h^n.$$

In particular, θ_h^n is nonnegative on $D_j \cup D$. For every $\varepsilon > 0$,

$$D_j \setminus D \subset \{|V_j - V| > \varepsilon\} \cup \{0 < h - V \leq \varepsilon\}.$$

Since $|\theta_h^n|$ is finite and absolutely continuous with respect to ω_X^n , $V_j \rightarrow V$ in capacity gives

$$\limsup_{j \rightarrow \infty} \int_{D_j \setminus D} \theta_h^n \leq \int_{\{0 < h - V \leq \varepsilon\}} |\theta_h^n|.$$

Letting $\varepsilon \downarrow 0$, we obtain $\int_{D_j \setminus D} \theta_h^n \rightarrow 0$. Hence, we see that

$$|\theta_{V_j}^n - \theta_V^n|(X) = \int_{D_j \triangle D} \theta_h^n = m_\phi - m_{\phi_j} + 2 \int_{D_j \setminus D} \theta_h^n \rightarrow 0,$$

where we use the fact that $m_{\phi_j} \rightarrow m_\phi$. □

4. CONSTRUCTION OF THE METRIC

We are now ready to prove Theorem 1.1. We divide the proof into several steps. *Step 1. Construction on $\mathcal{H}_{\theta, \phi}$* : Fix $f_0, f_1 \in C^{1,1}(X)$, and write

$$u_i^j = P_\theta[\phi_j](f_i), \quad u_i = P_\theta[\phi](f_i), \quad i = 0, 1.$$

Let u_t^j, u_t be the corresponding weak geodesics. We will prove

$$(4.1) \quad \lim_{j \rightarrow \infty} d_{p, \phi_j}(u_0^j, u_1^j)^p = \int_X |\dot{u}_0|^p \theta_{u_0}^n,$$

where the dot denotes the right derivative at $t = 0$. We follow the strategy in [Gup25, §4].

First, we choose $A > 0$ with $\theta \leq A\omega_X$. For $\tau \in \mathbb{R}$, by [DR16, Theorem 2.5], we have

$$h_\tau = P_{A\omega_X}(f_0, f_1 - \tau) \in C^{1,1}(X).$$

Since $h_\tau \leq f_0, f_1 - \tau$, $P_\theta[\phi_j](h_\tau) \leq P_\theta(u_0^j, u_1^j - \tau)$. Conversely, since $P_\theta(u_0^j, u_1^j - \tau) \in \text{PSH}(X, A\omega_X)$ lies below $f_0, f_1 - \tau$, we have $P_\theta(u_0^j, u_1^j - \tau) \leq h_\tau$. It also lies below $u_0^j \leq \phi_j + \sup_X f_0$. Hence

$$P_\theta(u_0^j, u_1^j - \tau) \leq P_\theta(\phi_j + \sup_X f_0, h_\tau) \leq P_\theta[\phi_j](h_\tau).$$

We conclude that $P_\theta[\phi_j](h_\tau) = P_\theta(u_0^j, u_1^j - \tau)$. Adapting the proofs of [Gup25, Lemmas 2.7 and 2.8] to endpoints with the same bounded relative singularity type, we obtain

$$\{\dot{u}_0^j \geq \tau\} = \{P_\theta(u_0^j, u_1^j - \tau) = u_0^j\} = \{P_\theta[\phi_j](h_\tau) = u_0^j\}$$

outside a pluripolar set. By [DT21], we know $\theta_{u_0}^n$ is supported on $D_j = \{u_0^j = f_0\}$. Thus, we obtain

$$\theta_{u_0}^n(\{\dot{u}_0^j \geq \tau\}) = \int_{\{P_\theta[\phi_j](h_\tau)=f_0\}} \theta_{f_0}^n = \theta_{P_\theta[\phi_j](h_\tau)}^n(\{h_\tau = f_0\}).$$

The last equality uses the almost everywhere equality of the Hessians of h_τ and f_0 on $\{h_\tau = f_0\}$. The same construction applied to ϕ gives the identity

$$\theta_{u_0}^n(\{\dot{u}_0 \geq \tau\}) = \int_{\{P_\theta[\phi](h_\tau)=f_0\}} \theta_{f_0}^n = \theta_{P_\theta[\phi](h_\tau)}^n(\{h_\tau = f_0\}).$$

Since $\{h_\tau = f_0\}$ is independent of j , Lemma 3.3 and the preceding identities give

$$\begin{aligned} & \left| \theta_{u_0}^n(\{\dot{u}_0^j \geq \tau\}) - \theta_{u_0}^n(\{\dot{u}_0 \geq \tau\}) \right| \\ & \leq \left| \theta_{P_\theta[\phi_j](h_\tau)}^n - \theta_{P_\theta[\phi](h_\tau)}^n \right|(\{h_\tau = f_0\}) \leq \left| \theta_{P_\theta[\phi_j](h_\tau)}^n - \theta_{P_\theta[\phi](h_\tau)}^n \right|(X) \rightarrow 0. \end{aligned}$$

Together with $m_{\phi_j} \rightarrow m_\phi$, this also gives convergence

$$\theta_{u_0}^n(\{\dot{u}_0^j < \tau\}) = m_{\phi_j} - \theta_{u_0}^n(\{\dot{u}_0^j \geq \tau\}) \longrightarrow m_\phi - \theta_{u_0}^n(\{\dot{u}_0 \geq \tau\}) = \theta_{u_0}^n(\{\dot{u}_0 < \tau\}).$$

Put $M := \|f_1 - f_0\|_\infty$. Since $|u_1^j - u_0^j| \leq M$, the geodesic barrier and convexity give

$$u_0^j - Mt \leq u_t^j \leq (1-t)u_0^j + tu_1^j \leq u_0^j + Mt.$$

Thus $|\dot{u}_0^j| \leq M$, and similarly $|\dot{u}_0| \leq M$. By the layer-cake formula,

$$\int_X |\dot{u}_0^j|^p \theta_{u_0^j}^n = p \int_0^M s^{p-1} \left[\theta_{u_0^j}^n(\{\dot{u}_0^j \geq s\}) + \theta_{u_0^j}^n(\{\dot{u}_0^j < -s\}) \right] ds.$$

The expression in brackets converges for every $s > 0$ and is bounded by m_{ϕ_j} , uniformly bounded in j . The dominated convergence theorem therefore gives

$$\int_X |\dot{u}_0^j|^p \theta_{u_0^j}^n \longrightarrow \int_X |\dot{u}_0|^p \theta_{u_0}^n.$$

By [Gup25, Theorem 3.10], the left-hand side equals $d_{p,\phi_j}(u_0^j, u_1^j)^p$. This proves (4.1). Since its right-hand side depends only on u_0, u_1 , the limit in (4.1) is independent of all choices.

Symmetry and the triangle inequality pass to the limit. Recall the I_p -energy:

$$I_p(u, v) := \int_X |u - v|^p (\theta_u^n + \theta_v^n).$$

Lemma 3.3 and the uniform bound $|u_0^j - u_1^j| \leq \|f_0 - f_1\|_\infty$ give

$$I_p(u_0^j, u_1^j) \rightarrow I_p(u_0, u_1).$$

Thus [Gup25, Lemma 3.9] yields a constant $C = C(n, p) > 0$ such that

$$(4.2) \quad C^{-1} I_p(u, v) \leq d_{p,\phi}(u, v)^p \leq C I_p(u, v), \quad u, v \in \mathcal{H}_{\theta,\phi}.$$

Non-degeneracy follows from the relative domination principle in [DDL25, Theorem 3.12].

Step 2. Extension to $\mathcal{E}^p(X, \theta, \phi)$: For $u \in \mathcal{E}^p(X, \theta, \phi)$, choose smooth functions $f_\ell \searrow u$. Then

$$u^\ell := P_\theta[\phi](f_\ell) \searrow u,$$

since $u \leq u^\ell \leq f_\ell$. By [Gup25, Theorem 2.1], I_p is continuous along simultaneous decreasing sequences. The argument of [Gup25, Theorem 4.11] therefore shows that

$$d_{p,\phi}(u, v) := \lim_{\ell \rightarrow \infty} d_{p,\phi}(u^\ell, v^\ell)$$

is well defined and independent of the approximations. As in [Gup25, Lemma 4.12 and Theorem 4.13], it is a metric, satisfies (4.2) on $\mathcal{E}^p(X, \theta, \phi)$, and is continuous along decreasing sequences.

Step 3. Completeness: Let (u_j) be $d_{p,\phi}$ -Cauchy. Hölder's inequality gives

$$I_1(u_j, u_k) \leq (2m_\phi)^{1-1/p} I_p(u_j, u_k)^{1/p} \rightarrow 0.$$

By [Gup23, Theorems 4.5 and 5.4], applied to the weight $|t|$, there is $u \in \mathcal{E}^1(X, \theta, \phi)$ such that $I_1(u_j, u) \rightarrow 0$ and $u_j \rightarrow u$ in capacity.

Here we also use I_p for the same nonnegative integral on $\text{PSH}(X, \theta)$, allowing the value $+\infty$. Fix $v \in \text{PSH}(X, \theta)$ and $A > 0$. The functions $\min\{|u_k - v|^p, A\}$ are nonnegative, uniformly bounded, and quasi-continuous and converge in capacity to $\min\{|u - v|^p, A\}$; values on pluripolar pole sets are immaterial. Applying [DDL25, Theorem 2.6] to $\theta_{u_k}^n$ and to the fixed measure θ_v^n gives

$$\int_X \min\{|u - v|^p, A\} (\theta_u^n + \theta_v^n) \leq \liminf_{k \rightarrow \infty} I_p(u_k, v).$$

Letting $A \rightarrow \infty$ and applying monotone convergence, we obtain

$$I_p(u, v) \leq \liminf_{k \rightarrow \infty} I_p(u_k, v).$$

Taking $v = \phi$ gives $I_p(u, \phi) \leq \sup_k I_p(u_k, \phi) < \infty$, where the uniform bound follows from (4.2) and the triangle inequality. Since u has full relative mass, it follows that $u \in \mathcal{E}^p(X, \theta, \phi)$. Taking $v = u_j$ and using the Cauchy property, we conclude that

$$d_{p,\phi}(u_j, u)^p \leq C I_p(u_j, u) \leq C \liminf_{k \rightarrow \infty} I_p(u_j, u_k) \rightarrow 0.$$

Step 4. Weak geodesics: The argument of [Gup25, Lemma 5.3] extends the endpoint formula to $u_0 \in \mathcal{H}_{\theta,\phi}$ and $u_1 \in \mathcal{E}^p(X, \theta, \phi)$:

$$d_{p,\phi}(u_0, u_1)^p = \int_X |\dot{u}_0|^p \theta_{u_0}^n.$$

Indeed, decreasing approximation of u_1 gives decreasing initial derivatives converging to $\dot{u}_0$. Their positive parts are uniformly bounded, while their negative parts increase. Dominated and monotone convergence, respectively, give the formula.

For $u_0, u_1 \in \mathcal{H}_{\theta,\phi}$, the slices u_t have the same singularity type as ϕ . Applying the endpoint formula to $s \mapsto u_{ts}$ and to the reversed geodesic gives

$$(4.3) \quad d_{p,\phi}(u_0, u_t) = t d_{p,\phi}(u_0, u_1), \quad d_{p,\phi}(u_t, u_1) = (1-t) d_{p,\phi}(u_0, u_1).$$

For arbitrary endpoints in $\mathcal{E}^p(X, \theta, \phi)$, choose $u_i^\ell \in \mathcal{H}_{\theta,\phi}$ decreasing to u_i ; then $u_i^\ell \searrow u_i$. By [DDL25, Corollary 3.19], $P_\theta(u_0, u_1) \in \mathcal{E}(X, \theta, \phi)$. The bounds $P_\theta(u_0, u_1) \leq u_t \leq \phi + C$ therefore imply $u_t \in \mathcal{E}(X, \theta, \phi)$. By (4.2), the triangle inequality, and (4.3),

$$I_p(u_t^\ell, \phi) \leq C (t d_{p,\phi}(u_0^\ell, u_1^\ell) + d_{p,\phi}(u_0^\ell, \phi))^p,$$

which is uniformly bounded by decreasing continuity at the endpoints. The truncation argument using [DDL25, Theorem 2.6] gives

$$I_p(u_t, \phi) \leq \liminf_{\ell \rightarrow \infty} I_p(u_t^\ell, \phi) < \infty.$$

Thus $u_t \in \mathcal{E}^p(X, \theta, \phi)$, and decreasing continuity extends (4.3) to arbitrary endpoints.

Finally, applying its second identity to $r \mapsto u_{rt}$, for $0 < s < t \leq 1$, yields

$$d_{p,\phi}(u_s, u_t) = \left(1 - \frac{s}{t}\right) d_{p,\phi}(u_0, u_t) = (t-s) d_{p,\phi}(u_0, u_1).$$

Hence weak geodesics are constant-speed metric geodesics.

Step 5. The case $p = 1$: For analytic singularity types, [Gup25, Lemmas 3.8 and 3.11] give the rooftop energy formula. For $u^j = P_\theta[\phi_j](f)$ and $u = P_\theta[\phi](f)$, Lemma 3.3, also applied to $h = 0$, gives $u^j \rightarrow u$ and $\phi_j \rightarrow \phi$ in capacity. Since $|u^j - \phi_j| \leq \|f\|_\infty$ and all mixed masses equal $m_{\phi_j} \rightarrow m_\phi$, we may apply [Gup25, Lemma 2.11] to the nonnegative weights $u^j - \phi_j + \|f\|_\infty$ and subtract the convergent constant terms to obtain

$$E_{\theta,\phi_j}(u^j) \longrightarrow E_{\theta,\phi}(u).$$

The same applies to the rooftops, using the obstacle h_0 above. The analytic energy formula therefore passes to the limit on $\mathcal{H}_{\theta,\phi}$. Decreasing approximation extends it to $\mathcal{E}^1(X, \theta, \phi)$, yielding (1.3). Here $P_\theta(u, v) = P_\theta[\phi](u, v)$, since $u, v \leq \phi + C$ for some constant C .

We also record the following useful inequality:

$$(4.4) \quad d_{p,\phi}(u, v)^p \leq \int_X (v - u)^p \theta_u^n, \quad u \leq v.$$

For endpoints in $\mathcal{H}_{\theta,\phi}$, this follows from $0 \leq \dot{u}_0 \leq v - u$ and (4.1). The general case follows by ordered decreasing approximations and the weighted integral convergence established in the proof of [Gup25, Theorem 2.1].

Finally, we end the note by comparing the metric $d_{p,\phi}$ on $\mathcal{E}^p(X, \theta, \phi)$ constructed above to the one constructed by Xia [Xia23], which we denote by $\hat{d}_{p,\phi}$. By [Xia23, Lemma 6.10], for $u \leq v$ it can be characterized as the infimum of the lengths of finite increasing chains joining u to v :

$$\hat{d}_{p,\phi}(u, v) = \inf_{u=u_0 \leq \dots \leq u_N=v} \sum_{i=0}^{N-1} \left(\int_X (u_{i+1} - u_i)^p \theta_{u_i}^n \right)^{1/p},$$

where all u_i belong to $\mathcal{E}^p(X, \theta, \phi)$. For any $u, v \in \mathcal{E}^p(X, \theta, \phi)$, set $P := P_\theta(u, v)$, which belongs to $\mathcal{E}^p(X, \theta, \phi)$ by [Xia23, Theorem 4.13]. The metric is given by the Pythagorean formula [Xia23, Definition 6.7(2)]:

$$\hat{d}_{p,\phi}(u, v)^p = \hat{d}_{p,\phi}(P, u)^p + \hat{d}_{p,\phi}(P, v)^p.$$

Corollary 4.5. *The metric $d_{p,\phi}$ is equivalent to Xia's metric $\hat{d}_{p,\phi}$, i.e., there exists $C = C(n, p) > 0$ such that*

$$C^{-1} d_{p,\phi}(u, v) \leq \hat{d}_{p,\phi}(u, v) \leq C d_{p,\phi}(u, v), \quad \text{for any } u, v \in \mathcal{E}^p(X, \theta, \phi).$$

In particular, $(\mathcal{E}^p(X, \theta, \phi), \hat{d}_{p,\phi})$ is complete.

Proof. For $u \leq v$, (4.4) and the triangle inequality give

$$d_{p,\phi}(u, v) \leq \hat{d}_{p,\phi}(u, v) \leq \left(\int_X (v - u)^p \theta_u^n \right)^{1/p}.$$

For any $u, v \in \mathcal{E}^p(X, \theta, \phi)$, we have from the triangle inequality and Hölder's inequality:

$$\begin{aligned} d_{p,\phi}(u, v) &\leq d_{p,\phi}(P_\theta(u, v), u) + d_{p,\phi}(P_\theta(u, v), v) \leq \hat{d}_{p,\phi}(P_\theta(u, v), u) + \hat{d}_{p,\phi}(P_\theta(u, v), v) \\ &\leq 2^{1-1/p} (\hat{d}_{p,\phi}(P_\theta(u, v), u)^p + \hat{d}_{p,\phi}(P_\theta(u, v), v)^p)^{1/p} = 2^{1-1/p} \hat{d}_{p,\phi}(u, v). \end{aligned}$$

Conversely, by [DDL25, Theorem 3.6], we have

$$\theta_{P_\theta(u, v)}^n \leq \mathbf{1}_{\{P_\theta(u, v)=u\}} \theta_u^n + \mathbf{1}_{\{P_\theta(u, v)=v\}} \theta_v^n.$$

Therefore, we have:

$$\hat{d}_{p,\phi}(u, v)^p \leq \int_X ((u - P_\theta(u, v))^p + (v - P_\theta(u, v))^p) \theta_P^n \leq I_p(u, v) \leq C d_{p,\phi}(u, v)^p.$$

Thus, $d_{p,\phi}$ and $\hat{d}_{p,\phi}$ are bi-Lipschitz equivalent and have the same Cauchy sequences. Completeness follows from Theorem 1.1. $\square$

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